Theoretical Probability

1 Consider this list of numbers:

$$2, 2, 2, 3, 3, 3, 4, 4, 5, 5, 5, 7, 7, 7, 7, 9, 9$$

a How many numbers are in the list?

b A number is chosen from the list at random. Find the probability that it is an odd number.

c State the number that has the same probability of being picked as 4.

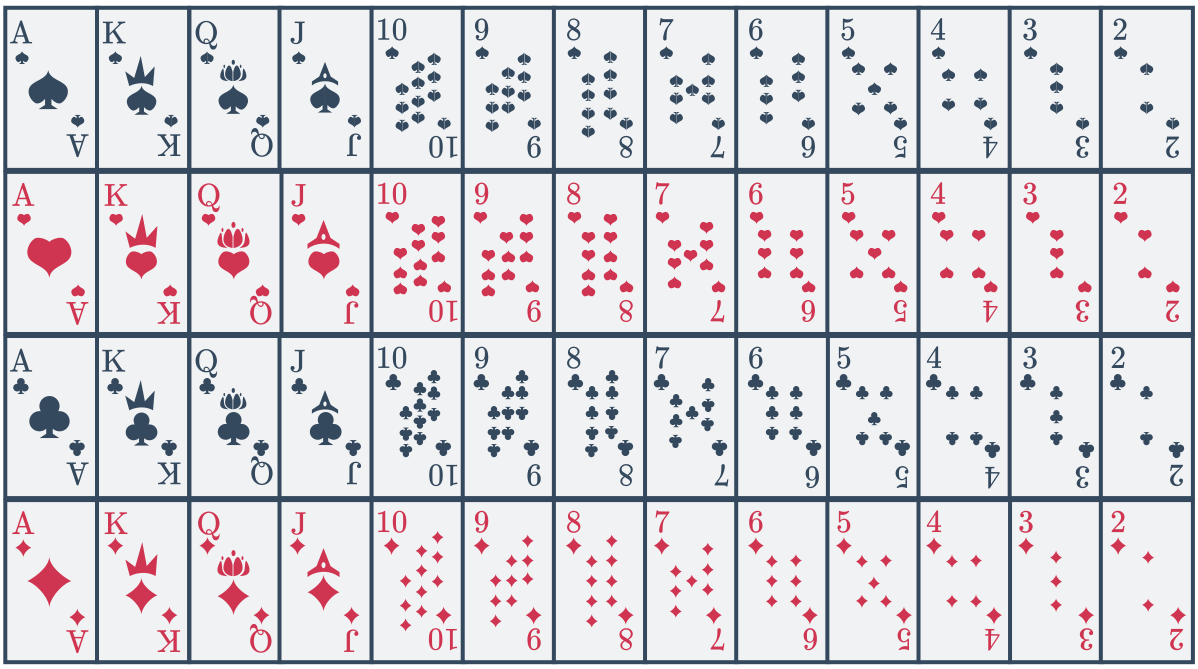
d A number is chosen from the list at random. State the number that has the highest probability of being chosen.

2 A number between 1 and 20 inclusive is randomly picked. Find:

a The probability that the number is at least 8.

b The probability that the number is less than 8.

3 A standard deck of 52 cards is shown below:



If a card is selected at random, find:

a The probability that it is a red card.

b The probability that it is a card between 5 and 9 inclusive.

c The probability that it is a card that is red and has a number between 5 and 9 inclusive.

d The probability that it is a card that is red or a king.

e The probability that it is a spade, with a number printed on it.

g The probability that it is a black or King[redacted] not both.

4 A spinner has 4 sectors with a star, 4 sectors with apple, and 2 sectors with an elephant. All the sectors are the same size.

a Find the probability of spinning an elephant.

b Find the probability of spinning an apple.

c Are the two events in parts (a) and (b) mutually exclusive?

d Are the two events in parts (a) and (b) complementary?

Probability of complimentary events

5 The probability that Victoria will win the major prize in a raffle at the school fete is 0.053. Find the probability that Victoria will not win.

6 The probability of the local football team winning their grand final is 0.36. Find the probability that they won't win the grand final.

7 The probability that it hails today is 0.43. Find the probability that it doesn't hail.

8 A biased coin is flipped, with heads and tails as possible outcomes. Calculate $P(\text{heads})$ if $P(\text{tails}) = 0.56$.

9 A company that makes sprockets guarantees that they will be within 0.5 mm either way of the client's chosen size. There is a probability of 0.968 that a sprocket will be within the allowable size, find the probability that a sprocket won't be within the allowable size.

10 A number between 1 and 100 inclusive is randomly picked. The probability that the number is less than 61 is $\dfrac{60}{100}$.

a What is the complement of drawing a number greater than 61?

b Find the probability that the number is greater than 61.

11 A number between 1 and 50 inclusive is randomly picked. The probability that the number is less than 42 is $\dfrac{82}{100}$.

a State the complement of drawing a number less than 42.

b State the complement of drawing a number that is at least 43.

c Find the probability that the number is at least 42.

12 The 26 letters of the alphabet are written on pieces of paper and placed in a bag. If one letter is picked out of the bag at random, find the probability of:

a Not selecting a B.

b Not selecting a C.

c Not selecting a K, R or T.

d Not selecting a K, L or M.

e Selecting a letter that is not in the word PROBABILITY.

f Not selecting a T, L, Q, A, K or Z.

g Not selecting a A, E, I, O or U.

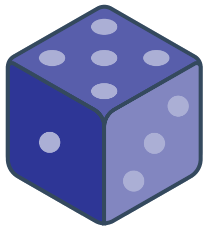
h Selecting a letter that is not in the word WORKBOOK.

13 A letter is chosen at random from the word ORDERED.

a State the letter that has the highest probability of not being chosen.

b Find the probability that the chosen letter is not "D".

14 A standard 6-sided die is rolled once.



a Find the probability of rolling a prime number.

b Find the probability of rolling an odd number.

c Are the two events in parts (a) and (b) mutually exclusive?

d Are the two events in parts (a) and (b) complementary?

e The probability of not rolling a 2.

f The probability of not rolling a 2 or 5.

g The probability of not rolling an odd number.

h The probability of not rolling a 9.

i The probability of not rolling a 1, 2, 3, 4, 5 or 6.

15

a A bag contains red marbles and blue marbles. Given the probability of drawing a red marble is $21/44$, find the probability of drawing a blue marble.

b Another bag contains 34 red marbles a[redacted] blue marbles. If a marble is picked at random, find:

i $P(\text{red})$ **ii** $P(\text{Not Red})$

16 A bag contains 50 black marbles, 37 orange marbles, 29 green marbles and 23 pink marbles. If a marble is selected at random, find the following probabilities, in simplest form:

a P(orange) **b** P(orange or pink)

c P(not orange) **d** P(neither orange nor pink)

17 Mario has a bag of marbles. It contains 3 white marbles and 5 marbles of other colours. Mario picks a marble from the bag without looking.

a Find the probability that Mario picks a white marble.

b Find the probability that Mario picks a marble that is not white.

c Are the two events in parts (a) and (b) mutually exclusive?

d Are the two events in parts (a) and (b) complementary?

18 A marble is chosen at random from a box containing 4 different colour marbles, red, blue, green and purple. The probability of selecting different colours is given in the table. Find:

a The probability of not selecting a green marble.

b The probability of not selecting a blue marble.

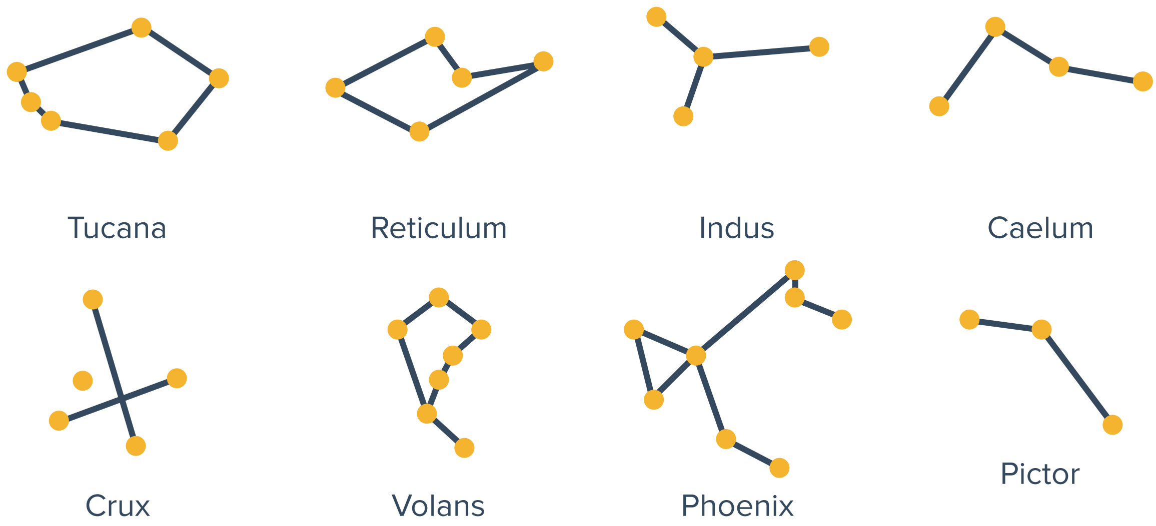
c The probability of selecting a red or purple marble.

d The probability of not selecting a purple marble.

e The probability of selecting neither a blue or green marble.

Colour	Probability
Red	$\frac{3}{13}$
Blue	$\frac{2}{9}$
Green	$\frac{1}{5}$

19 A constellation is randomly selected from [redacted]ght shown below:



a Find the probability that the name of the constellation begins with a vowel.

b The complementary event is selecting a constellation that begins with a consonant. Find the probability of this event.

c Find the probability that the constellation has 6 or more stars.

d The complementary event is selecting a constellation that has less than 6 stars. Find the probability of this event.